SPE-226971-MS

Integration of Diagenetic Rock Typing with Advanced Geophysical Attributes for Well Placement Optimization in Underbalanced Coiled Tubing Drilling

Abdulaziz M. Alrashed, Dhahran, Saudi Arabia / Saudi Aramco; Taher M. Sodagar, Dhahran, Saudi Arabia / Saudi Aramco; Weihua Wang, Dhahran, Saudi Arabia / Saudi Aramco

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Underbalanced Coiled Tubing Drilling (UBCTD), which is the practice of drilling in a wellbore pressure that lower than the formation static pressure to allow the well to flow while drilling, became a common practice in Saudi Arabia in recent years due to its ability to significantly enhance wells performance in terms of production and drilling efficiency. However, common drawbacks of this technique are the lack of real time data due to the slim hole size as it only captures the gamma-ray and gas rate while drilling, and it is highly affected by the subsurface heterogeneities and diagenesis of the targeted reservoir such as clay precipitations and facies changes. Therefore, optimum well placement is required to improve well results which can be achieved through integration of different techniques such as Diagenetic Rock Typing and advanced geophysical attributes such as Mono-frequency, RMS-amplitude and acoustic impedance. Diagenetic Rock Typing is a technique which separates the reservoir into different rock types based on core, log, and petrographic analysis of the clastic reservoir in which the best rock type shows the least amount of total clay content, typically less than 5%, and highest effective porosity, commonly more than 5%. These rock types can be propagated into a 3D domain by applying geostatistical techniques to map subsurface heterogeneities. Integration of this techniques with other geophysical attributes including Mono-frequency, which is a technique used to identify gas sweet spots by separating the seismic volume into different frequency bands, results in generating a map that supports optimum placement for UBCTD wells.

The feasibility of this techniques was tested in a clastic reservoir. The reservoir is known for its heterogeneity and sever diagenetic processes such as clay precipitation, quartz cementation, and quartz overgrowth. An anomaly was identified in field "A" in an area where there is minimal wells penetration. The anomaly was identified as a potential gas bearing reservoir using Diagenetic Rock Typing, Mono-frequency with other seismic attributes including RMS amplitudes. Two UBCTD wells, 'Well-1' and 'Well-2', were drilled targeting the sweet spot shown in the various maps, and produced exceptional results. Further UBCTD operations can be enhanced, in terms of production and economic feasibility, using similar approaches through integration of different reservoir characterization solutions.

Introduction

Underbalanced Coiled Tubing Drilling (UBCTD) has emerged as an innovative technology in the development of the complex subsurface reservoirs offering various advantages such as minimizing formation damage, improved drilling efficiency, and enhanced production. Underbalanced condition is achieved by maintaining the hydrostatic pressure to be lower than the reservoir pressure which allows the well to flow while drilling (Saponja, 1998). Although this technique comes with many advantages that help in enhanced recovery and operational efficiency, it has major drawbacks such as limited real time geosteering and monitoring tools due to slim hole size conditions (Abu Alsaud, 2024). Therefore, the success of UBCTD heavily depend on the optimal well placement, more precisely in heterogenous reservoirs where diagenesis and variations in porosity and permeability present significant challenges.

The study area is focused on complex clastic reservoir. It is one of the most prolific hydrocarbon basins in the world for both clastic and carbonate reservoirs. The targeted reservoir is composed of clastic heterogeneous sandstone which is known for its structural complexity and severe diagenetic process such as clay precipitation and quartz overgrowth. Diagenetic rock typing, which separates the reservoir into different rock types based on the variations of their porosity and total clay, along with various geophysical attributes such as Mono-frequency, were employed to assess the feasibility for potential gas sweet spot in an area with minimal wells penetration. The significance of this study is that it will be used to identify sweet spot area for UBCTD potential. To attain and validate the objectives of this study, one field is used where 2 UBCTD wells showed exceptional results by flowback tests in the clastic reservoir.

Geological Background

The study area is mainly composed of clastic reservoir within sedimentary succession. The reservoir showed a range of porosity and permeability in the field wells with various reservoir thicknesses due to the structural complexity. During the time of the deposition, the Arabian Plate lay at high southern latitudes which is approximately a time equivalent of the glacial deposits recognized in Oman (Konert, et al., 2001) Recent paleotectonic global reconstructions (Figure 1) show that the northernmost extent of the Gondwanan ice cap in late Carboniferous times in the most southerly areas of the Arabian Plate placing it in a period of high glaciation event which heavily influenced the deposition of the targeted reservoir. For instance, clastics of the targeted reservoir were deposited in a continental to marginal environments during a period of glaciation and laid down upon an unconformity making the thickness vary widely from zero to hundreds of feet (Al-Masrahy, 2020, Al-Masrahy 2024). The structural framework of the reservoir was highly influenced by the tectonic history of the Arabian plate.

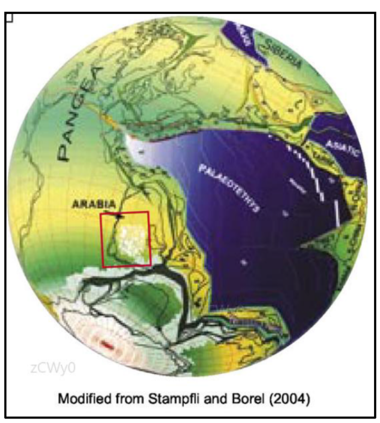

Figure 1—This diagram shows the location of the Arabian plate during late carboniferous, showing the glaciation event which contributed to the sedimentation of the targeted reservoir.

The targeted reservoir has been deposited in glaciofluvial environments according to Al-Masrahy, (2024). Two distinct units were recognized, an upper coastal aeolian facies and lower fluvial facies. The upper unit is composed of high to low angle cross beds of medium sandstones along with finer rippled sandstone with shale laminae. The lowers unit is massive fins grained, well sorted sandstones with sheetflood and proglacial stream deposits accompanied with coarse sandstones. Operationally, it is often recognized with slow ROP and high degree of quartz overgrowth and cementation.

The reservoir petrophysical properties have been significantly altered due to diagenetic processes contributing to its server complexity and heterogeneities. Early diagenesis including grain replacing and pore-filling clays such as chlorite, kaolinite, and illite, where chlorite being the most abundant clay mineral, ranging from 0-5%, in the targeted reservoir. Late diagenesis of the target begins with the precipitation of quartz overgrowths and illite cementation. These diagenetic processes along with the original grain size sorting play the major controls on the prosody-permeability relations in the study area. Coarser sandstones with more quartz cement show much higher permeability and porosity than finer, illite cemented sandstones.

The geological difficulty of the targeted reservoir shown by the structural complexity, faulting, fractures, early and late diagenetic processes emphasizes on the need of reservoir characterization solutions for Under balanced coiled tubing (UBCTD) operations. Diagenetic Rock Typing plays a major role in identifying the vertical and lateral facies changes caused by the diagenesis process and thus mitigate the risk associated with the UBCTD. Integration of such technique with other geophysical attributes such Mono-frequency, RMS amplitude, and acoustic impedance, is essential in mapping potential gas sweet spots in the targeted reservoir for optimized UBCTD wells placement.

Methodology

The methodology of this workflow is to integrate Diagenetic Rock Typing and Mono-frequency along with various geophysical attributes to construct a map for potential gas sweet spots for optimized well placement for Under Balanced Coil Tubing wells within the highly heterogeneous and complex clastic reservoir. This workflow leverages multidisciplinary data sources including well log data, 3D seismic, and petrographic analysis to characterize the targeted reservoir. The goal of this approach is to produce a robust map by combining the diagenetic rock typing and Mono-frequency for sweet spot mapping and place the laterals with the best section using the DRT 3D model to ensure significant outcomes tailored to UBCTD operational challenges.

Diagenetic Rock Typing Approach

Various sedimentological and petrographic studies on sandstones show that it is mainly composed of quartz and minor feldspars as grain contents, cements including quartz overgrowth, and other clay minerals such as illite, kaolinite and chlorite. Petrographic analysis was conducted on the targeted clastic reservoir to showcase the various clay minerals present as showed in figures 2A–2F where different examples of plane polarized light thin section as well as various samples of scanning electric microscope (SEM) images showing different sandstone textures and their main components including framework grains (quartz) and authigenic minerals such as illite, kaolinite, and chlorite. Various diagenetic processes discussed affected the sandstones and provided rigid foundation for wireline log based petrophysical evaluation in terms of the sandstone mineralogy. Other diagenetic processes which affect the sandstones intergranular porosity, and make it become tighter are compaction and cementation.

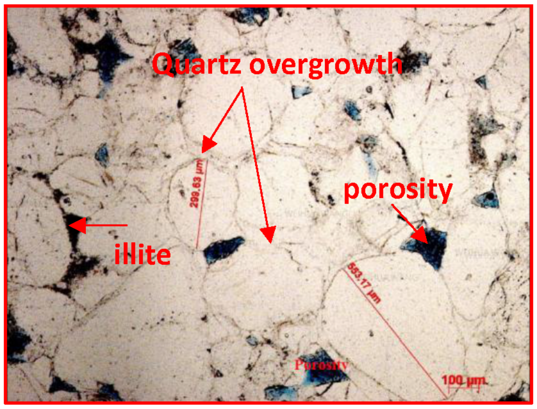

Figure 2A—This section image showing sandstone with quartz overgrowth, porosity and illite.

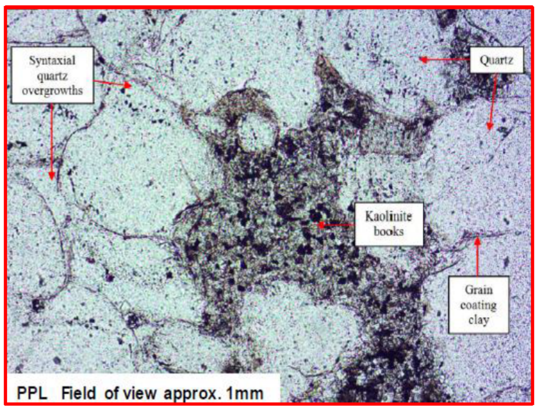

Figure 2B—This section image showing sandstone with abundant quartz grain, quartz overgrowth, grain coating clay and booklet kaolinite.

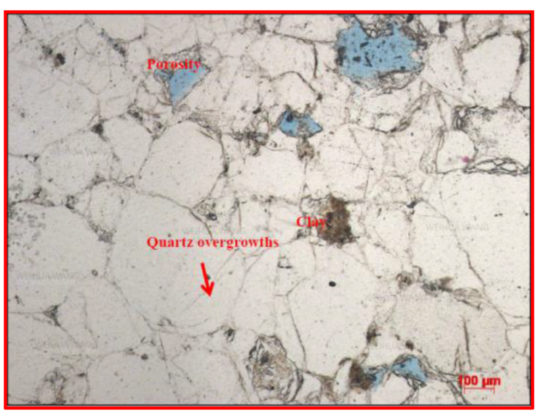

Figure 2C—This section image showing sandstone with abundant quartz overgrowth, some clay and less porosity.

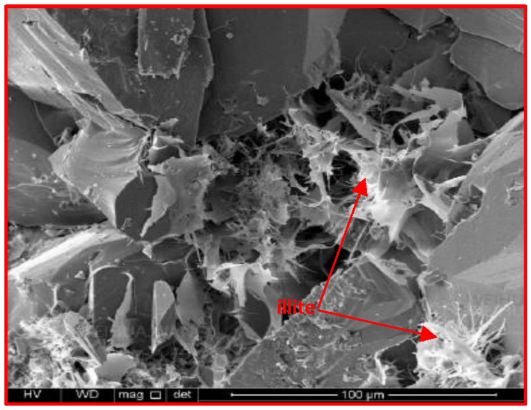

Figure 2D—SEM image showing sandstone with abundant illite

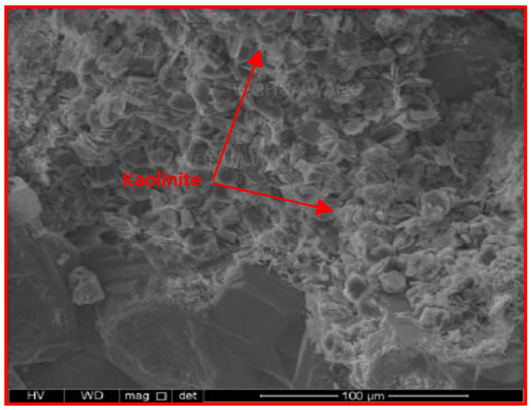

Figure 2E—SEM image includes sandstone with abundant pore filling booklet kaolinite.

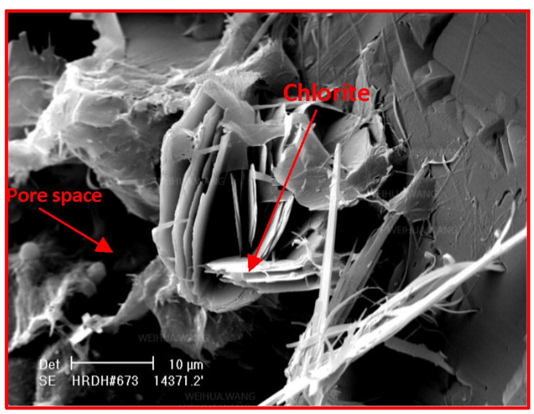

Figure 2F—SEM image includes sandstone with pore space and minor chlorite.

Diagenetic rock typing is a new approach which utilizes existing petrophysical evaluating to categorize rocks into distinct diagenetic types based on differences in total clay content and effective porosity. (Wang, 2024). Two main diagenetic process that contribute to the reduction of intergranular porosity in sandstones Compaction and cementation. Compaction is mainly a function of burial depth, which increases heavily with more burial depth resulting in more intense compaction. On the other hand, cementation plays a more significant role for the same sandstone as it is the primary reason for the sand heterogonies. This is because different types and amounts of cements can significantly impact the porosity and permeability of the sandstone. For example, clean sandstones with quartz cement tend to have higher porosity and permeability compared to sandstones rich in clay minerals. Notably, even when porosity is similar, quartz-cemented sandstones often have better permeability. As a result, different diagenetic rock types are identified based on their total clay content, including the presence different clay minerals such as chlorite, kaolinite, and illite. This is why the inventor, (Wang, 2024), has utilized total clay content as one of the key factors in performing sandstone diagenetic rock typing.

The diagenetic rock typing criteria is established based on a comprehensive sandstone petrography study, which considers various factors such as the total clay content. For instance, the total clay content is less than 5% in clean sandstones. Moreover, the under balanced coil tubing drilling in clean sandstone with less than 5% clay can naturally produce gas (Wang, 2024). Furthermore, sandstones, based on various analysis of sandstone frackability, with more than 3% porosity and less than 10% total clay content can have significant impact on production after utilizing hydraulic fracturing (Wang, 2024). Therefore, based on the various petrographic and petrophysical analysis, sandstones can be divided into six different type of diagenetic rock types as shown in Table A-1 and Figure 3 as follows: TYPE1, TYPE2, TYPE3, TYPE4, TYPE5 and TYPE6. Lastly, these rock types are distributed in 3D domain which allows the laterals to be placed in the optimum type.

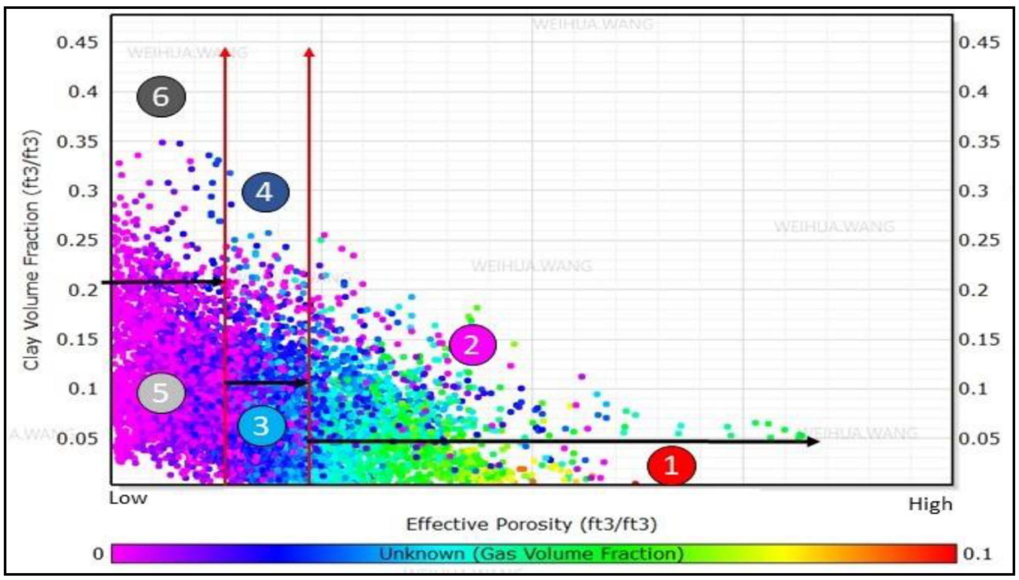

Figure 3—This diagram showing the tight sandstone PHIE vs total clay cross plot, and color change reflect gas volume variation in the tight sunstone. The studied tight sandstone can be divided into six different diagenetic rock types based on their effective porosity and total clay variations.

Mono-Frequency Ratio Approach

Geophysical data processors have usually decomposed and attenuated the seismic signals such as, low-frequency ground roll, 50- or 60-Hz cultural noise, and high-frequency random noise into its Fourier (or spectral) components. Furthermore, they have enhanced the signal through different geophysical techniques including seismic deconvolution and wavelet-shaping (Chopra and Marfurt, 2007). Conventional methods for identifying fluid response are based on Gassman's theory, which assumes frequency-independent seismic effects. Moreover, studies have demonstrated that spectral decomposition can be a valuable tool for extracting insights into fluid properties, offering a more enhanced approach to fluid detection (Castagna et al., 2003; Ebrom, 2004; Odebeatu et al., 2006, Zhang et al., 2008).

To further analyze seismic data, researchers have developed techniques to calculate frequency-dependent seismic reflection responses and properties related to frequencies, such as attenuation and dispersion. Spectral decomposition methods, as described by Mallat and Zhang (1993), are particularly effective in extracting frequency-related attributes from seismic data, allowing for the decomposition of seismograms into constituent wavelets and the construction of time-frequency analyses. This enables a more detailed understanding of the seismic data and the properties of the subsurface.

In areas with gas reservoirs, which are often associated with amplitude anomalies, low-frequency shadow is frequently observed, where the seismic frequencies appear to be lowered beneath the reservoir. These low-frequency shadows occur because of the attenuation of the high-frequency energy in the gas column within the reservoir. This effect is more pronounced in thicker gas reservoirs, where the energy has a longer travel path, and can be used as an indicator of the presence of hydrocarbons (Chopra and Marfurt, 2007).

The primary objective of Mono Frequency Spectral Decomposition (FSD) methods is to evaluate the reliability of spectral information as a Direct Hydrocarbon Indicator (DHI) (Sodagar, 2014, 2015, 2018 and 2019). Another significant application is to recognize the geologic features highlighted by spectral decomposition (Chopra and Marfurt, 2007). Other applications of the Mono-frequency ratio include enhancing the visibility of stratigraphic features, improved resolution, estimating the thickness of thin beds, reducing noise, and optimizing spectral balancing (Castagna et al., 2003).

Mono-frequency spectral decomposition is the process to separate seismic data into several frequency bands (Sodagar, 2014, 2015, 2018 and 2019). This technique allows generating frequency ratio maps of decomposed low to high frequency of targeted reservoir interval to detect the gas fluid type since gas column attenuates at higher seismic frequencies for the same reservoir level (Sodagar, 2014, 2015, 2018 and 2019). Before applying the Mono FSD Ratio Map approach, it is essential to apply a Relative Amplitude Preserved (RAP) processing workflow to the 3D seismic data, including multiple and noise removals, to obtain an original, unmodified, and high Signal-to-Noise (S/N) ratio seismic dataset (Sodagar, 2014, 2015, 2018, and 2019). This step ensures that the preservation of seismic data in its original form, with minimal modifications, and has a high signal-to-noise ratio, which is crucial for accurate interpretation and analysis using the Mono FSD Ratio Map approach. The workflow as implemented by Sodagar is obtained by generating the instantaneous frequency and dominant frequency volumes from the processed 3D seismic data where the lower and higher frequencies are clearly observed on these frequency volumes at the reservoir level. After that, decomposing the 3D seismic volume to mono frequency spectra of low to high volumes with increment of 10 Hz, then extracting the average amplitudes between the top and the bottom of the interested reservoir interval for each Mono FSD volumes, which create Mono FSD maps for the low to the high frequencies for the same reservoir intervals. Finally, dividing the low frequency maps to the high frequency maps to get the ratio maps at the targeted reservoir. This allows generating a sweet spots map for potential gas reservoirs for optimum UBCTD wells placement. The frequency ratio map for this workflow was obtained by dividing the 10 Hz over 40 Hz to generate sweet spot for potential gas.

Discussion and Results

Generating thickness maps for the best rock types in the diagenetic rock typing (DRT) technique (Type 1, Type 2, and Type 3), and combining them with the Mono-Frequency Ratio map, allows for producing a sweet spot map to locate the best gas potential area in terms of rock types and seismic data. This technique allows for the optimum well placement for UBCTD. Additionally, the well path design and lateral targeting was optimized using the diagenetic rock typing 3D model. This approach was tested using two UBCTD wells in an area with minimal wells penetration where these wells were placed in the best surface locations based on the DRT, Mono-Frequency Ratio, and the combined map as shown in Figure 4, Figure 5, and Figure 6 respectively. Additionally, the laterals were planned and placed in the best rock types as shown in the DRT cross section in Figure 7. The two UBCTD wells showed excellent performance and flowed great gas rates while drilling through the best rock types predicted by the DRT 3D model. Due to the optimum wells performance while drilling, the planned lateral footage was cut short since the objective of the wells were met. Additionally, the flowback test after drilling showed exceptional performance where these wells showed the highest flowback rates in the field for the targeted reservoir. As a result of the excellent performance, the planned two laterals for both wells were dropped, leading to an exceptional cost saving. The findings from this integrated workflow of combining these two attributes has produced a robust workflow that significantly enhanced wells placement within the targeted clastic reservoir, highlighting the high potential of this approach to address and overcome the challenges and limitations associated with UBCTD wells and operations.

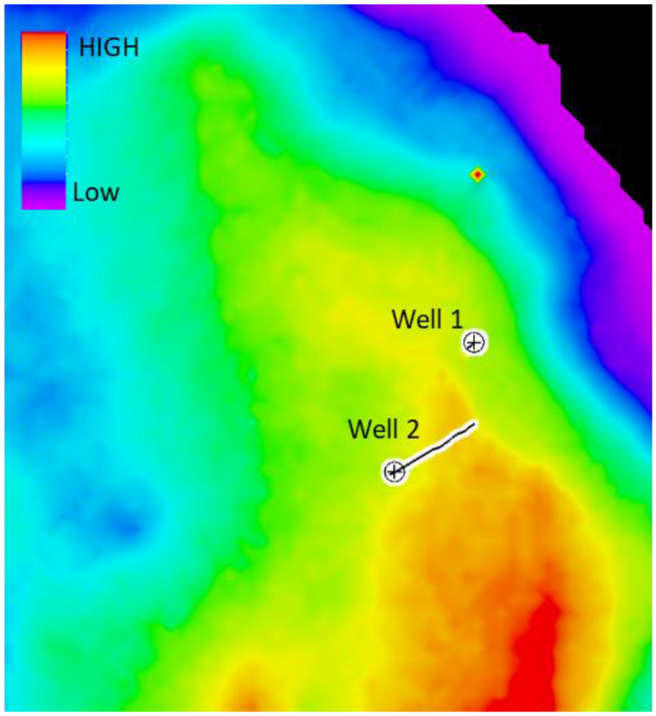

Figure 4—This figure shows the wells location on the thickness map of the best rock types (Type 1, Type 2, and Types 3) from the Diagenetic rock typing model.

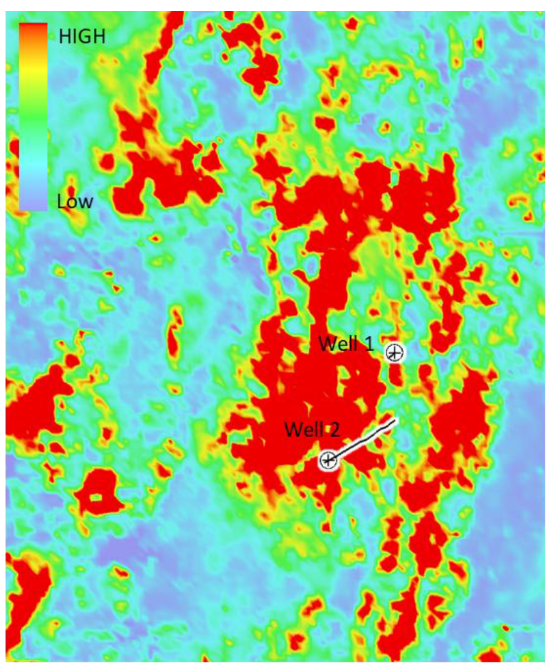

Figure 5—This figure shows the wells location on the mono-frequency ratio map obtained from analyzing the seismic data. The frequency ration used is 10/40 Hz.

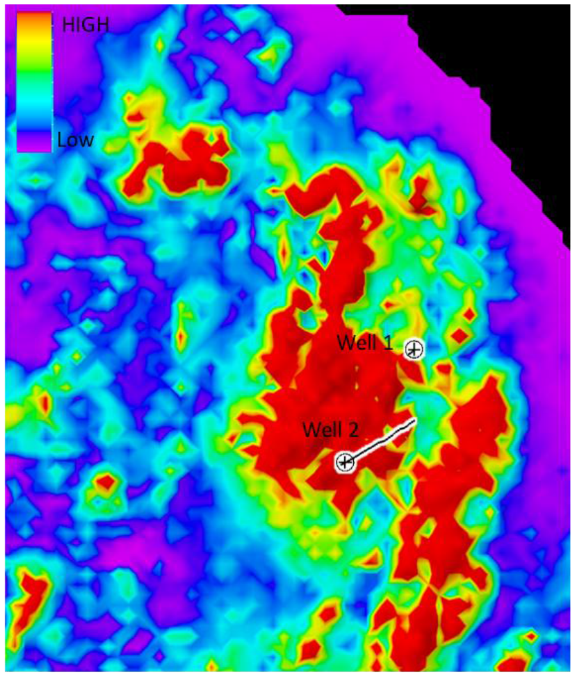

Figure 6—This figure shows the wells location on the combined map obtained by merging the optimum rock types from DRT and the mono-frequency ratio map.

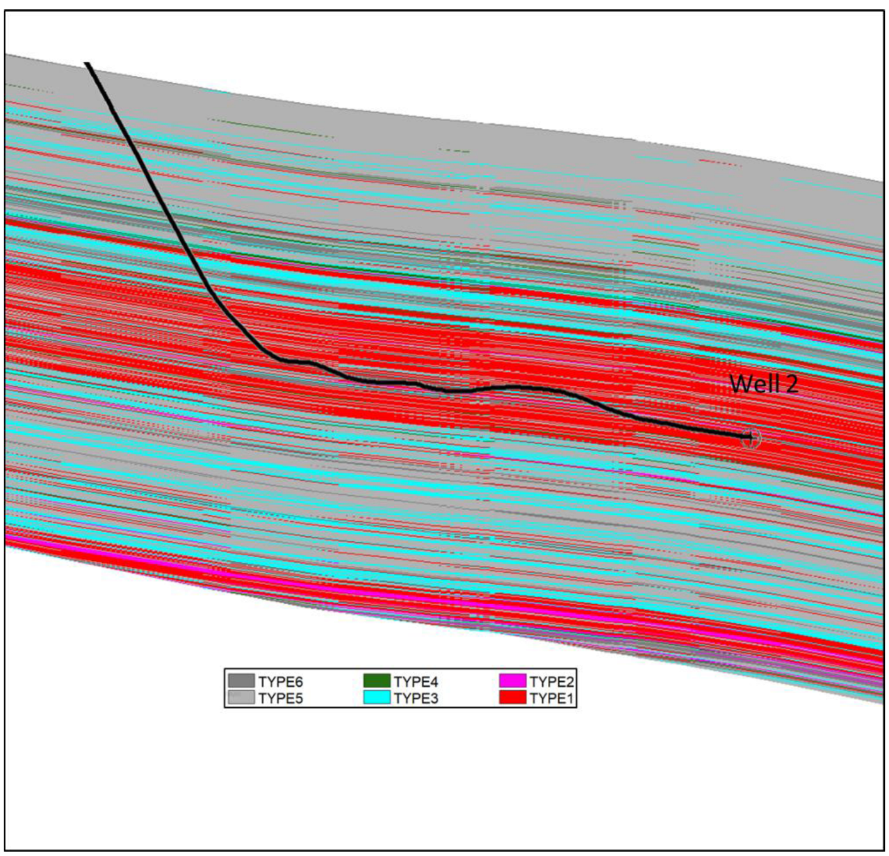

Figure 7—The figure shows the cross-section of Well-2 which targeted the best rock type.

Table 1—Classification of tight sandstone diagenetic rock typing for fracturing.

Diagenetic rock type	Code	PHIE	Total clay
Type 1	6	>5%	<5%
Type 2	5	>5%	*≥5%*
Type 3	4	3%≤ and ≤5%	<10%
Type 4	3	3%≤ and ≤5%	*≥10%*
Type 5	2	<3%	<20%
Type 6	1	<3%	*≥20%*

Conclusion

The study developed an innovative workflow integrating diagenetic rock typing with mono-frequency ratio to optimize well placement in under balanced coil tubing drilling within a heterogeneous clastic reservoir. The tectonic history of the Arabian plate starting from the Hercynian orogeny to the sedimentation of the targeted reservoir heavily contributed to the structural complexity. Furthermore, early and late diagenetic processes such as compaction and cementation led to the sever heterogeneities of the clastic reservoir. These diagenetic processes contributed to the accumulation of different clay minerals, which heavily disturbed the vertical and lateral continuity of the reservoir. The methodology addressed these reservoir challenges by combining a detailed petrophysical and geophysical analysis, interpretation, and mapping. The diagenetic rock typing focused on the petrophysical analysis by using multiple datasets such as well logs, and petrographic thin sections, to characterized the diagenetic rocks into 6 different types where the best type can flow gas naturally while drilling since it is having more than 5% effective porosity and less than 5% total clay content. These rock types were propagated across and modeled in 3D across the field. On the other hand, the mono-frequency approach was used to provide geophysical analysis and interpretation in the study area. It was obtained by decomposing an original, unmodified, and high Signal-to-Noise (S/N) ratio 3D seismic volume to mono frequency spectra of low to high volumes with increment of 10 Hz. The ratio used in this workflow is 10/40 Hz because it showed indication of gas presence. Integrating both maps produced a sweet spot map that indication the best locations for gas presence with the best rock types. This approach was tested with two UBCTD wells which were drilled in an area with minimal wells penetration. The wells were planned targeting the sweet spots from DRT, Mono-Frequency Ratio Map and merged map, and they showed exceptional results while drilling and after obtaining the flowback test.

This workflow enhances reservoir characterization, reduces operational risks associated with diagenesis processes, and improves economic outcomes in UBCTD by enabling optimized wells placements. This approach is scalable to other clastic reservoirs on a similar stratigraphic setting. Further enhancement of such approach can be done by incorporating other reservoir characterization and real time geosteering technologies.

Acknowledgments

The authors would like to especially thank all their colleagues and the reviewers for their extreme valued efforts and valuable time, precious guidance, valuable comments, recommendations, and bright suggestions that helped to significantly improve this study and subsequent paper.

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